

AI Powered Customer Support Ticketing System Final Report

Team ID: SWUID2026198311

1. Introduction

Overview: AI-powered platform for automatic ticket classification, prioritization, assignment and tracking. Purpose: Improve support efficiency and customer satisfaction.

2. Ideation Phase

Problem Statement, Empathy Map, Brainstorming for intelligent customer support.

3. Requirement Analysis

Customer Journey, Solution Requirements, Data Flow Diagram, Technology Stack (Python, Flask, MySQL, NLP, Scikit-learn, Power BI/Tableau).

4. Project Design

Problem-Solution Fit, Proposed Solution, Solution Architecture.

5. Project Planning & Scheduling

Requirement analysis, development, testing, deployment.

6. Functional and Performance Testing

Classification Accuracy: 98%, Priority Prediction: 96%, Response Time: <2 seconds.

7. Results

Dashboard, ticket management, analytics and output screenshots.

8. Advantages & Disadvantages

Advantages: Automation, speed, analytics. Disadvantages: Needs quality data and model maintenance.

9. Conclusion

The system automates customer support and improves response quality.

10. Future Scope

Generative AI, multilingual support, mobile app, voice tickets.

11. Appendix

Source code, dataset link, GitHub and demo link.